

Day 14

M T W T F S

24/28/24

11:00

⇒ Correlation:- Correlation is a statistical technique used to determine the degree to which two variables are related.

For example:- the price and demand of a product is in correlation.

➔ Types of correlation:- 1- Positive correlation -

2- Negative correlation

Positive correlation:- It is a relation between two variables when one variable increases the other one also increases.

Negative correlation:- It is a relation between two variables when one variable decreases the other one increases.

⇒ The Correlation Coefficient:- It is a strength of the linear relationship between two variables.

This measurement is between -1 and +1.
and The Correlation is denoted by r .

-1 Strong
Negative
correlation

0
No
correlation

+1 Strong
Positive
correlation

Correlation Coefficient formula:-

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n-1) s_x s_y}$$

⇒ Problem

AGE (X)	Glucose (Y)
42	98
23	68
22	73
47	79
50	88
60	82

Scatter Plots:-

The standard way to visualize the relation ship between two measured data variable. The x-axis represent one variable and y-axis another and each point on the graph is a record.

